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ACADEMIC DETAILS

Career objective:

I am a Ph.D. research scholar in Biotechnology with strong expertise in oral cancer biology, clinical sample handling, molecular techniques, and cancer bioinformatics. Experienced in RNA extraction, cDNA synthesis, qRT-PCR, microbiome analysis, and in-silico drug discovery. Co-inventor of a Government of India–granted patent and author of peer-reviewed international publications. Seeking research positions in academia or research organizations aligned with biomedical and molecular research.

Academic Qualification:

- **Ph.D. (Biotechnology), 2022–Present, Amity University Lucknow Campus, Thesis Submitted**
- M. Sc 2018-20 Dr. Shakuntala Mishra National Rehabilitation University (Microbiology)
- B. Sc 2015-18 Veer Bahadur Singh Purvanchal University (Zoology, Botany, Chemistry)
- Intermediate 2013-15 Saraswati Vidya Mandir (CBSE)
- High School 2011-13 Sir Syed Ahmad Public School (CBSE)

Professional Experience:

Work Experience:

- **Teaching experience:** Contributed to UG and PG Biotechnology teaching through lectures, laboratory practicals, lab supervision, and examination invigilation (January 2024 to April 2025) at Amity Institute of Biotechnology, Amity University Uttar Pradesh, Lucknow Campus.
- Experienced in **clinical sample handling** and **molecular biology techniques**, with hands-on expertise in **oral cancer patient saliva samples**. Proficient in clinical sample collection under approved **ethical clearance (IEC 20/24)**, **total RNA extraction**, **cDNA synthesis**, and **quantitative real-time PCR (qRT-PCR)**. I am skilled in qRT-PCR data analysis and interpretation for gene expression studies. Actively involved in **wet-lab research** at Dr. Ram Manohar Lohia Institute of Medical Sciences (DR. RMLIMS), Lucknow, adhering strictly to biosafety and ethical guidelines.

Training/Workshop Attend

- One-day workshop on Rational Drug Development using Artificial Intelligence and Insilco Methods held at Amity University Uttar Pradesh, Lucknow, on 4th November 2022.
- DST-STUTI Sponsored Hands-on Training Program on “Advanced Molecular Biology Techniques” organized by the Department of Bioscience & IIRC, Integral University Lucknow, from 1st December to 7th December 2022.
- A Workshop on “Introduction to Microbiome/Metagenome Data Analysis” was organized and conducted by the Bioinformatics facility Rajiv Gandhi for Biotechnology (RGCV) on 25th February 2023.
- Two Days Workshop on “Computer Aided Drug Design Using BIOVIA Discovery Studio Suite”

organized by the Department of Bioengineering in collaboration with the Integral Startups Foundation, under the Skill Development Initiative, held on 18-19 September 2024 at Integral University, Lucknow.

- 3-hour online workshop titled “Multi-Ligand Docking and ADMET Analysis” organized by Research Boulevard Technologies on October 14, 2024.
- “One-week Hands-on Industrial Skill Training Program with State-of-the-Art Lectures on HPLC, PCR, RT-PCR, Primer Designing & ELISA Reader” organized by MRD LifeScience, Lucknow, dated from October 04-10, 2025.

Achievement:

- "Co-inventor of a patented '**Non-Invasive Oral Cancer Test Device**' (Design No. **434669-001, Government of India**)."
- Received the **Best Oral Presentation Award** at the 4th International Sustainability Conference on Health, Safety, Fire, and Environmental Advance (HSFEA-2023), held on October 30–31, 2023.
- Awarded the **Travel Grant Award** for participation in the 4th International Sustainability Conference on Health, Safety, Fire, and Environmental Advance (HSFEA-2023), held on October 30–31, 2023.

List of Papers Published

A. Research Paper

- **Singh, N. K.**, Awasthi, P., Gupta, A., Anand, N., Giri, B. S., & Hasan, S. (2025). Environmental Hazards and Chemoresistance in OTSCC: Molecular Docking and Prediction of Paclitaxel and Imatinib as BCL2 and EGFR Inhibitors. *Biology*, 14(9), 1174. <https://doi.org/10.3390/biology14091174>
- **Singh, N. K.**, Awasthi, P., Gupta, A., Anand, N., Giri, B. S., Hassan, M. I., & Hasan, S. Identifying *Porphyromonas gingivalis*-infected hub genes and molecular mechanisms of oral squamous cell carcinoma pathogenesis. *Discov Appl Sci* 7, 163 (2025). <https://doi.org/10.1007/s42452-024-06260-y>
- Hasan, S., **Singh, N. K.**, Awasthi, P., & Gupta, A. (2024). TARGETING ROLES OF TCF3 AND WNT SIGNALING PATHWAY IN HPV POSITIVE HEAD AND NECK SQUAMOUS CELL CARCINOMA: BIOINFORMATICS ANALYSIS. *Science and innovation*, 3(Special Issue 45), 94-101.

B. Book Chapter

- Awasthi, P., **Singh, N. K.**, Gupta, A., Anand, N., & Hasan, S. (2025). AI and deep learning in cancer genomics. In *Deep Learning in Genetics and Genomics* (pp. 97-123). Academic Press.
- INTRODUCTION TO CANCER BIOLOGY AND GENETICS: Cancer Genetics Unlocked. (2024). Notion Press.

Computer Skills

- Experienced in bioinformatics databases and tools (NCBI, TCGA, STRING, Cytoscape, DAVID, Enrichr, Metascape, molecular docking tools), Cancer Bioinformatics
- Proficient in MS Word, Excel, and PowerPoint

Training/Conferences/Symposium

- Attend a two-day workshop on “Computer Aided Drug Design Using BIOVIA Discovery Studio Suite” organized by the Department of Bioengineering in Integral University on September 18-19, 2024.
- Attend virtual International Conference in Uzbekistan, “Status and Development Prospects of Fundamental and Applied Microbiology: The Viewpoint of Young Scientists” on September 25-26, 2024.
- Attending online workshop titled “Multi-Ligand Docking and ADMET Analysis,” organized by Research

Boulevard Technologies on October 14, 2024.

- 4th International Sustainability Conference on Health, Safety, Fire, and Environmental advances (HSFEA 2023) October 30-31, 2023
- International Conference on High- Throughput Data Analysis and Intelligence in Biological Research at Amity University Lucknow Campus. October 19-20, 2023
- A Seminar on Lifestyle Diseases, Working Women's Health Priorities and Interventions. September 23, 2023
- Lecture Series Cum One Week Faculty Development Program (LSFDP 1.0), 2023
- 4th International Sustainability Conference on Health, Safety, Fire, and Environmental advances (HSFEA 2023) October 30-31, 2023
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- Lecture Series Cum One Week Faculty Development Program (LSFDP 1.0), 2023
- 4th International Sustainability Conference on Health, Safety, Fire, and Environmental advances (HSFEA 2023) October 30-31, 2023
- International Conference on High- Throughput Data Analysis and Intelligence in Biological Research at Amity University Lucknow Campus. October 19-20, 2023.
- A Seminar on Lifestyle Diseases Working Women's Health Priorities and Interventions. September 23, 2023
- Lecture Series Cum One Week Faculty Development Program (LSFDP 1.0), 2023
- 14th National Conference on Science and Technology Rural Development, Participate and Present a Paper (Poster) Organized by The Indian Science Congress Association (Allahabad Chapter) DSMNRU Lucknow.

Skills:

- Ability to work in a team, and a strong work ethic.
- Communication, enthusiasm, and teamwork.
- Sincere and self-confident.

Hobbies:

- Reading scientific literature
- Academic writing
- Playing badminton
- Travelling

Declaration:

I hereby declare that the above-mentioned information is correct to my knowledge and I bear the responsibility for the correctness of the above-mentioned information.



(Nishant Kumar Singh)